

HR Schema Business Insights

Using JOINS and WHERE Clause

Tank54 Group | Oracle HR Schema

This guide shows what business insights you can get from the Oracle HR database. Each insight answers a real business question. We use INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN, and WHERE clause to get the answers.

All examples use the Oracle HR schema tables: EMPLOYEES, DEPARTMENTS, LOCATIONS, JOBS, JOB_HISTORY, COUNTRIES, and REGIONS.

The Oracle HR Schema

The Oracle HR schema is a sample database that stores information about a company. It has tables for employees, departments, jobs, locations, and more. Here are the main tables we will use and what they contain:

Table	What It Stores	Key Columns
EMPLOYEES	All workers in the company	employee_id, first_name, last_name, salary, department_id, manager_id, hire_date, job_id, commission_pct
DEPARTMENTS	Company departments	department_id, department_name, manager_id, location_id
JOBS	Job titles and salary ranges	job_id, job_title, min_salary, max_salary
LOCATIONS	Office locations	location_id, street_address, city, country_id
JOB_HISTORY	Past jobs of employees	employee_id, job_id, start_date, end_date, department_id
COUNTRIES	Country names and region	country_id, country_name, region_id
REGIONS	World regions	region_id, region_name

How Tables Connect:

- EMPLOYEES connects to DEPARTMENTS by DEPARTMENT_ID
- EMPLOYEES connects to JOBS by JOB_ID
- EMPLOYEES connects to EMPLOYEES (self) by MANAGER_ID (manager is also an employee)
- DEPARTMENTS connects to LOCATIONS by LOCATION_ID
- LOCATIONS connects to COUNTRIES by COUNTRY_ID
- COUNTRIES connects to REGIONS by REGION_ID
- JOB_HISTORY connects to EMPLOYEES and JOBS by EMPLOYEE_ID and JOB_ID

Insight 1: Employee Full Profile

Business Question:

What is the full profile of each employee? I want to see the employee name, job title, department name, and the city where they work. This helps HR see the complete picture of every employee in one row.

SQL Query:

```
SELECT e.first_name,
       e.last_name,
       j.job_title,
       d.department_name,
       l.city
FROM employees e
INNER JOIN departments d
    ON e.department_id = d.department_id
INNER JOIN jobs j
    ON e.job_id = j.job_id
INNER JOIN locations l
    ON d.location_id = l.location_id
ORDER BY d.department_name, e.last_name;
```

How It Works:

We need data from 4 tables: EMPLOYEES (name), JOBS (job title), DEPARTMENTS (department name), and LOCATIONS (city). We use 3 INNER JOINS to connect them. The chain is: EMPLOYEES --department_id--> DEPARTMENTS --location_id--> LOCATIONS, and EMPLOYEES --job_id--> JOBS. We use INNER JOIN because we only want employees who have a department, a job, and a location. This gives HR a clean list of all active employees with their full information in one place.

Sample Result:

First Name	Last Name	Job Title	Department	City
Steven	King	President	Executive	Seattle
Neena	Kochhar	Administration Vice President	Executive	Seattle
Diana	Lorentz	IT Programmer	IT	South San Francisco

Insight 2: Employees Without a Department

Business Question:

Are there any employees who are NOT assigned to any department? This is important for HR because every employee should belong to a department. If someone has no department, this is a data problem that needs to be fixed.

SQL Query:

```
SELECT e.first_name,
       e.last_name,
       e.job_id,
       e.hire_date
FROM employees e
LEFT JOIN departments d
    ON e.department_id = d.department_id
WHERE d.department_id IS NULL;
```

How It Works:

We use LEFT JOIN because we want ALL employees, even those without a department. LEFT JOIN keeps all rows from EMPLOYEES. If an employee has no matching department, the department columns will be null. Then we add WHERE d.department_id IS NULL. This filters the result and shows ONLY the employees where the department is null. In other words: "Show me only the employees who have no department." This is a data quality check. If this query returns rows, HR knows that some employees are missing a department assignment.

Sample Result:

First Name	Last Name	Job ID	Hire Date
Kimberely	Grant	SA_REP	24-MAY-2007

Insight 3: Departments With No Employees

Business Question:

Are there any departments that have ZERO employees? This helps management see if there are departments that were created but nobody works in them. Maybe the department is new, or maybe all employees left and nobody noticed.

SQL Query:

```
SELECT d.department_name,  
       d.location_id  
FROM departments d  
LEFT JOIN employees e  
      ON d.department_id = e.department_id  
WHERE e.employee_id IS NULL;
```

How It Works:

We want ALL departments, so we start from DEPARTMENTS (left table) and use LEFT JOIN to EMPLOYEES. LEFT JOIN keeps all departments. If a department has no employees, the employee columns will be null. WHERE e.employee_id IS NULL filters and shows only departments where no employee was found. This is useful because an empty department means the company is paying for a location and a manager but has no staff. Management can decide what to do with these departments.

Sample Result:

Department Name	Location ID
Payroll	1700
Treasury	1700
Corporate Tax	1700
Control And Credit	1700
Shareholder Services	1700
Benefits	1700

Insight 4: Data Quality: Missing Connections

Business Question:

Show me ALL employees and ALL departments, even if they do not match. I want to see which employees have no department AND which departments have no employees. This is a full data quality report.

SQL Query:

```
SELECT e.first_name,  
       e.last_name,  
       d.department_name  
FROM employees e  
FULL OUTER JOIN departments d  
      ON e.department_id = d.department_id  
WHERE e.employee_id IS NULL  
      OR d.department_id IS NULL;
```

How It Works:

FULL OUTER JOIN shows everything from both tables. Matched rows appear normally. Unmatched employees show null for department. Unmatched departments show null for employee. We add WHERE e.employee_id IS NULL OR d.department_id IS NULL. This shows ONLY the problem rows: employees with no department, and departments with no employees. This is the most complete data quality check because it finds problems on both sides in one query. HR and IT can use this report to fix data issues.

Sample Result:

First Name	Last Name	Department Name
Kimberely	Grant	null
null	null	Payroll
null	null	Treasury

Insight 5: Department Headcount

Business Question:

How many employees work in each department? Management needs to know the size of each team to plan resources and compare team sizes across the company.

SQL Query:

```
SELECT d.department_name,  
       COUNT(e.employee_id) AS total_employees  
FROM departments d  
LEFT JOIN employees e  
      ON d.department_id = e.department_id  
GROUP BY d.department_name  
ORDER BY total_employees DESC;
```

How It Works:

We start from DEPARTMENTS and use LEFT JOIN to EMPLOYEES. This way, even departments with zero employees will appear in the result. COUNT(e.employee_id) counts the number of employees in each department. GROUP BY d.department_name puts employees into groups by department. ORDER BY total_employees DESC shows the biggest departments first. We use LEFT JOIN (not INNER JOIN) because we want to see empty departments too (they will show 0 employees). This report helps management understand how people are distributed across the company and which departments are overstaffed or understaffed.

Sample Result:

Department Name	Total Employees
Shipping	45
Sales	34
IT	5
Executive	3
Payroll	0

Insight 6: High Salary Employees by Department

Business Question:

Who are the highest paid employees in each department? Show employees who earn more than 10000, with their department name and job title. This helps management understand the salary distribution and see who the top earners are.

SQL Query:

```
SELECT e.first_name,
       e.last_name,
       e.salary,
       j.job_title,
       d.department_name
FROM employees e
INNER JOIN departments d
      ON e.department_id = d.department_id
INNER JOIN jobs j
      ON e.job_id = j.job_id
WHERE e.salary > 10000
ORDER BY e.salary DESC;
```

How It Works:

We join 3 tables: EMPLOYEES (name and salary), DEPARTMENTS (department name), and JOBS (job title). We use INNER JOIN because we only want employees who have a department and a job. WHERE e.salary > 10000 is the filter. It keeps only employees whose salary is more than 10000. ORDER BY e.salary DESC shows the highest salary first. This report is useful for budget planning. Management can see how many employees are in the high salary range and in which departments they work.

Sample Result:

First Name	Last Name	Salary	Job Title	Department
Steven	King	24000	President	Executive
Neena	Kochhar	17000	Administration VP	Executive
Lex	De Haan	17000	Administration VP	Executive
Alexander	Hunold	9000	IT Manager	IT

Insight 7: Employees and Their Manager

Business Question:

Show each employee and their manager name. Every employee (except the CEO) has a manager. The MANAGER_ID column in EMPLOYEES points to another employee. This helps HR see the reporting structure of the company.

SQL Query:

```
SELECT e.first_name AS employee_first,
       e.last_name AS employee_last,
       m.first_name AS manager_first,
       m.last_name AS manager_last
FROM employees e
LEFT JOIN employees m
      ON e.manager_id = m.employee_id
ORDER BY m.last_name, e.last_name;
```

How It Works:

This is a SELF JOIN. We join the EMPLOYEES table with itself. The first copy ("e") is for the employee. The second copy ("m") is for the manager. ON e.manager_id = m.employee_id means: "Find the manager by matching the employee manager_id with another employee employee_id". We use LEFT JOIN because the CEO (Steven King) has no manager. With LEFT JOIN, the CEO will appear with null for manager name. AS is used to give columns clear names (employee_first, manager_first). This report shows the full reporting chain of the company.

Sample Result:

Employee First	Employee Last	Manager First	Manager Last
Steven	King	null	null
Neena	Kochhar	Steven	King
Lex	De Haan	Neena	Kochhar
Alexander	Hunold	Neena	Kochhar

Insight 8: Employees Earning Outside Salary Range

Business Question:

Are there employees whose salary is outside the expected range for their job? Each job has a minimum and maximum salary. If an employee earns less than the minimum or more than the maximum, this is a problem. This is a salary compliance check.

SQL Query:

```
SELECT e.first_name,
       e.last_name,
       e.salary,
       j.job_title,
       j.min_salary,
       j.max_salary
FROM employees e
INNER JOIN jobs j
  ON e.job_id = j.job_id
WHERE e.salary < j.min_salary
      OR e.salary > j.max_salary;
```

How It Works:

We join EMPLOYEES with JOBS to get the salary range for each job. The JOBS table has min_salary and max_salary for each job title. WHERE e.salary < j.min_salary OR e.salary > j.max_salary filters for employees whose salary is below the minimum OR above the maximum. If this query returns rows, it means some employees have a salary problem. For example, someone might be paid less than they should be, or someone might be paid more than the allowed maximum. HR uses this report for salary audits and to fix pay issues.

Sample Result:

First Name	Last Name	Salary	Job Title	Min Salary	Max Salary
Kimberely	Grant	7000	Sales Representative	6000	12000

Insight 9: Employees by Location and Country

Business Question:

How many employees work in each city? And which country is each city in? This helps management understand where the company workforce is located. It is useful for planning office space and deciding where to hire more people.

SQL Query:

```
SELECT l.city,
       c.country_name,
       COUNT(e.employee_id) AS employee_count
FROM locations l
LEFT JOIN departments d
  ON l.location_id = d.location_id
LEFT JOIN employees e
  ON d.department_id = e.department_id
INNER JOIN countries c
  ON l.country_id = c.country_id
GROUP BY l.city, c.country_name
ORDER BY employee_count DESC;
```

How It Works:

We need 4 tables: LOCATIONS (city), COUNTRIES (country name), DEPARTMENTS (bridge between location and employee), and EMPLOYEES. We start from LOCATIONS and use LEFT JOIN to DEPARTMENTS, then LEFT JOIN to EMPLOYEES. This way, even locations

with no employees will appear. INNER JOIN to COUNTRIES because every location should have a country. COUNT(e.employee_id) counts how many employees are in each city. GROUP BY l.city, c.country_name groups the count by city and country. This report shows management where people work and which offices are empty.

Sample Result:

City	Country	Employee Count
Seattle	United States of America	3
South San Francisco	United States of America	5
Oxford	United Kingdom	1

Insight 10: Employee Job Changes

Business Question:

Which employees changed their job or department in the past? Show their old job, old department, new job, and when the change happened. This helps HR see the career path of employees and understand internal movements in the company.

SQL Query:

```
SELECT e.first_name,
       e.last_name,
       jh.job_id AS old_job,
       d.department_name AS old_department,
       jh.start_date,
       jh.end_date
FROM job_history jh
INNER JOIN employees e
      ON jh.employee_id = e.employee_id
LEFT JOIN departments d
      ON jh.department_id = d.department_id
ORDER BY e.last_name, jh.start_date;
```

How It Works:

We start from JOB_HISTORY because this table records past job changes. We join to EMPLOYEES to get the employee name. We join to DEPARTMENTS to get the old department name. We use LEFT JOIN for departments in case the old department was deleted. The result shows each job change with the old job, old department, and the date range (start_date to end_date). This report is useful for HR to see which employees moved between departments, who got promoted, and what career paths people took. It also helps in workforce planning and talent management.

Sample Result:

First Name	Last Name	Old Job	Old Department	Start Date	End Date
Jennifer	Whalen	AD_ASST	Administration	17-SEP-2003	24-MAR-2006
Neena	Kochhar	ST_MAN	IT	21-SEP-2005	24-MAR-2006

Insight 11: Employees With Commission

Business Question:

Which employees earn a commission? Show their name, salary, commission percent, department, and total pay (salary + commission). Sales employees often earn commission on top of their salary. Management wants to see who earns commission and how much.

SQL Query:

```
SELECT e.first_name,
       e.last_name,
       e.salary,
       e.commission_pct,
       d.department_name,
       ROUND(e.salary + (e.salary * e.commission_pct), 2)
       AS total_pay
FROM employees e
INNER JOIN departments d
      ON e.department_id = d.department_id
WHERE e.commission_pct IS NOT NULL
ORDER BY total_pay DESC;
```

How It Works:

We join EMPLOYEES with DEPARTMENTS to get the department name. WHERE e.commission_pct IS NOT NULL is the key filter. In the HR database, most employees have null commission_pct. Only sales employees have a commission percent. This filter keeps only employees who earn commission. ROUND(e.salary + (e.salary * e.commission_pct), 2) calculates the total pay. It adds the salary plus the

commission amount. ROUND with 2 shows 2 decimal places. ORDER BY total_pay DESC shows the highest total pay first. Management uses this to see the total cost of sales employees, including their commissions.

Sample Result:

First Name	Last Name	Salary	Commission %	Department	Total Pay
Janette	King	10000	0.35	Sales	13500
Patrick	Sully	9500	0.35	Sales	12825
Allan	McEwen	9000	0.35	Sales	12150

Insight 12: New Hires by Department

Business Question:

Which employees were hired in the last year? Show their name, department, job title, and hire date. This helps HR track new employees and see which departments are growing and hiring new people.

SQL Query:

```
SELECT e.first_name,
       e.last_name,
       e.hire_date,
       j.job_title,
       d.department_name
FROM employees e
INNER JOIN departments d
      ON e.department_id = d.department_id
INNER JOIN jobs j
      ON e.job_id = j.job_id
WHERE e.hire_date >= ADD_MONTHS(TRUNC(SYSDATE), -12)
ORDER BY e.hire_date DESC;
```

How It Works:

We join 3 tables: EMPLOYEES, DEPARTMENTS, and JOBS. WHERE e.hire_date >= ADD_MONTHS(TRUNC(SYSDATE), -12) is the filter. SYSDATE is the current date. ADD_MONTHS(..., -12) goes back 12 months. TRUNC removes the time part and keeps only the date. So this filter means: "Show employees hired in the last 12 months." ORDER BY e.hire_date DESC shows the newest hires first. HR uses this report for onboarding, training planning, and to see which departments are actively hiring.

Sample Result:

First Name	Last Name	Hire Date	Job Title	Department
Kimberely	Grant	24-MAY-2007	Sales Rep	Sales
Douglas	Grant	13-JAN-2008	Sales Manager	Sales

Insight 13: Average Salary by Job Title

Business Question:

What is the average, minimum, and maximum salary for each job title? Compare these numbers with the expected salary range for each job. This helps HR see if the company pays above or below the market range.

SQL Query:

```
SELECT j.job_title,
       j.min_salary AS range_min,
       j.max_salary AS range_max,
       ROUND(AVG(e.salary)) AS avg_paid,
       MIN(e.salary) AS actual_min,
       MAX(e.salary) AS actual_max
FROM jobs j
LEFT JOIN employees e
      ON j.job_id = e.job_id
GROUP BY j.job_title, j.min_salary, j.max_salary
ORDER BY avg_paid DESC;
```

How It Works:

We start from JOBS because we want to see all job titles, even those with no employees. LEFT JOIN to EMPLOYEES. AVG(e.salary) calculates the average salary for each job. MIN and MAX show the actual lowest and highest salaries paid. We compare these with j.min_salary and j.max_salary (the expected range). GROUP BY groups all employees by their job title. ROUND formats the average to a whole number. This report helps HR understand if salaries are fair and within range. If the average is much lower than the range, employees may be underpaid.

Sample Result:

Job Title	Range Min	Range Max	Avg Paid	Actual Min	Actual Max
President	20080	40000	24000	24000	24000
Administration VP	15000	30000	17000	17000	17000
IT Programmer	4000	10000	5760	4200	9000
Sales Representative	6000	12000	8350	6100	11500

Insight 14: Departments by Region

Business Question:

How many departments are in each region (Americas, Europe, Asia)? And how many employees work in each region? This gives management a global view of the company structure.

SQL Query:

```
SELECT r.region_name,
       COUNT(DISTINCT d.department_id) AS dept_count,
       COUNT(e.employee_id) AS emp_count
FROM regions r
INNER JOIN countries c
    ON r.region_id = c.region_id
LEFT JOIN locations l
    ON c.country_id = l.country_id
LEFT JOIN departments d
    ON l.location_id = d.location_id
LEFT JOIN employees e
    ON d.department_id = e.department_id
GROUP BY r.region_name
ORDER BY emp_count DESC;
```

How It Works:

This query uses 5 tables connected in a chain. REGIONS --region_id--> COUNTRIES --country_id--> LOCATIONS --location_id--> DEPARTMENTS --department_id--> EMPLOYEES. We start from REGIONS because we want to see all regions. We use LEFT JOIN for the rest so that regions with no locations still appear. COUNT(DISTINCT d.department_id) counts how many different departments are in each region. DISTINCT is important because multiple employees in the same department should count as only 1 department. COUNT(e.employee_id) counts total employees. GROUP BY r.region_name groups everything by region. This gives management a global overview: which regions are big, which are small, and where the workforce is concentrated.

Sample Result:

Region	Departments	Employees
Americas	6	50
Europe	2	8
Asia	1	2

Insight 15: Managers and Their Team Size

Business Question:

Show each manager, their department, and how many people report to them. Include managers who have zero direct reports. This helps HR see which managers have large teams and which have no team.

SQL Query:

```
SELECT e.first_name AS manager_first,
       e.last_name AS manager_last,
       d.department_name,
       COUNT(w.employee_id) AS team_size
FROM employees e
INNER JOIN departments d
    ON e.employee_id = d.manager_id
LEFT JOIN employees w
    ON e.employee_id = w.manager_id
    AND w.employee_id != e.employee_id
GROUP BY e.first_name, e.last_name, d.department_name
ORDER BY team_size DESC;
```

How It Works:

This query finds all managers. A manager is an employee whose employee_id appears in the manager_id column of the DEPARTMENTS table. INNER JOIN departments ON e.employee_id = d.manager_id finds only employees who are department managers. LEFT JOIN

employees w ON e.employee_id = w.manager_id finds all people who report to this manager. "w" is the worker (team member). w.employee_id != e.employee_id is important: it makes sure the manager is not counted as their own team member. COUNT(w.employee_id) counts direct reports. GROUP BY groups by manager name and department. Managers with no team will show 0. This report is useful for HR to see the span of control for each manager and to check if workloads are balanced.

Sample Result:

Manager First	Manager Last	Department	Team Size
Shelley	Higgins	Accounting	3
Neena	Kochhar	IT	5
Steven	King	Executive	2

Quick Reference

Here is a summary of all insights and which SQL techniques they use:

#	Insight	JOIN Type	WHERE Clause	Tables Used
1	Employee Full Profile	3x INNER JOIN	No	EMP, DEPT, JOB, LOC
2	Employees Without Department	LEFT JOIN	Yes (IS NULL)	EMP, DEPT
3	Departments With No Employees	LEFT JOIN	Yes (IS NULL)	DEPT, EMP
4	Data Quality: Missing Links	FULL OUTER JOIN	Yes (IS NULL + OR)	EMP, DEPT
5	Department Headcount	LEFT JOIN	No (GROUP BY)	DEPT, EMP
6	High Salary Employees	2x INNER JOIN	Yes (salary >)	EMP, DEPT, JOB
7	Employees and Their Manager	SELF LEFT JOIN	No	EMP, EMP
8	Salary Outside Range	INNER JOIN	Yes (OR, < , >)	EMP, JOB
9	Employees by Location	2x LEFT + 1x INNER	No (GROUP BY)	LOC, DEPT, EMP, CNT
10	Employee Job Changes	INNER + LEFT JOIN	No	JH, EMP, DEPT
11	Employees With Commission	INNER JOIN	Yes (IS NOT NULL)	EMP, DEPT
12	New Hires by Department	2x INNER JOIN	Yes (date)	EMP, DEPT, JOB
13	Average Salary by Job	LEFT JOIN	No (GROUP BY)	JOB, EMP
14	Departments by Region	1x INNER + 3x LEFT	No (GROUP BY)	REG, CNT, LOC, DEPT, EMP
15	Managers and Team Size	INNER + SELF LEFT	No (GROUP BY)	EMP, DEPT, EMP

Key SQL Patterns Used

Pattern	When to Use	Example
INNER JOIN + WHERE	You want matched rows AND you need to filter them	Employees with salary > 10000
LEFT JOIN + WHERE IS NULL	Find rows in the left table with NO match in the right table	Employees without a department
FULL OUTER JOIN + WHERE IS NULL	Find unmatched rows from BOTH tables	Data quality: missing connections
LEFT JOIN + GROUP BY	Count or sum grouped data, including empty groups	Department headcount (including 0)
SELF JOIN	A table connects to itself (manager = employee)	Employee and their manager
Multiple JOINS (3+)	You need data from more than 2 tables	Employee profile with 4 tables

Remember:

- JOIN connects tables. WHERE filters rows. Use them together for powerful queries.
- IS NULL finds empty or missing values. IS NOT NULL finds rows that have a value.
- LEFT JOIN + WHERE IS NULL = find orphans (rows with no match).
- GROUP BY + COUNT/AVG/SUM/MIN/MAX = get numbers and totals per group.